

Solutions: Linear Algebra

ECON 441: Introduction to Mathematical Economics

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Exercise 4.2

$$1. \ A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix} \quad C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$$

$$(a) \ A + B = \begin{bmatrix} 7 & 3 \\ 9 & 7 \end{bmatrix}$$

$$(b) \ C - A = \begin{bmatrix} 1 & 4 \\ 0 & -8 \end{bmatrix}$$

$$(c) \ 3A = \begin{bmatrix} 21 & -3 \\ 18 & 27 \end{bmatrix}$$

$$(d) \ 4B + 2C = \begin{bmatrix} 0 & 16 \\ 12 & -8 \end{bmatrix} + \begin{bmatrix} 16 & 6 \\ 12 & 2 \end{bmatrix} = \begin{bmatrix} 16 & 22 \\ 24 & -6 \end{bmatrix}$$

$$2. \ A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix} \quad C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$$

- (a) AB is defined as number of columns in A is two which is equal to the number of rows in B .

$$AB = \begin{bmatrix} 2 \times 2 + 8 \times 3 & 2 \times 0 + 8 \times 8 \\ 3 \times 2 + 0 \times 3 & 3 \times 0 + 0 \times 8 \\ 5 \times 2 + 1 \times 3 & 5 \times 0 + 1 \times 8 \end{bmatrix} = \begin{bmatrix} 28 & 64 \\ 6 & 0 \\ 13 & 8 \end{bmatrix}$$

Not possible to calculate BA as B has two columns, but A has three rows.

- (b) BC and CB are both defined as both have two rows and two columns.

$$BC = \begin{bmatrix} 14 & 4 \\ 69 & 30 \end{bmatrix} \neq CB = \begin{bmatrix} 20 & 16 \\ 21 & 24 \end{bmatrix}$$

$$4. \quad (a) \quad \begin{bmatrix} 0 & 2 \\ 36 & 20 \\ 16 & 3 \end{bmatrix}_{3 \times 2}$$

$$(b) \quad \begin{bmatrix} 49 & 3 \\ 4 & 3 \end{bmatrix}_{2 \times 2}$$

$$(c) \quad \begin{bmatrix} 3x + 5y \\ 4x + 2y - 7z \end{bmatrix}_{2 \times 1}$$

$$(d) \quad [7a + c \quad 2b + 4c]$$

Exercise 4.4

5. (e)

$$A = \begin{bmatrix} -2 \\ 4 \\ 7 \end{bmatrix}_{3 \times 1} \quad B = \begin{bmatrix} 3 & 6 & -2 \end{bmatrix}_{1 \times 3}$$

$$C = AB = \begin{bmatrix} -6 & -12 & 4 \\ 12 & 24 & -8 \\ 21 & 42 & -14 \end{bmatrix}_{3 \times 3}$$

$$D = BA = [3 \times -2 + 6 \times 4 + -2 \times 7] = [4]_{1 \times 1}$$

7. In example 5, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}$. In which case,

$$\begin{aligned} x^T A x &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \\ &= \begin{bmatrix} a_{11}x_1 & a_{22}x_2 \end{bmatrix}_{1 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \\ &= \underbrace{a_{11}x_1^2 + a_{22}x_2^2}_{\text{Weighted sum of squares}} = \sum_{i=1}^2 a_{ii}x_i^2 \end{aligned}$$

So $x^T A x$ represents a weighted sum of squares where a_{11}, a_{22} are weights.

But now what if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. In this case,

$$\begin{aligned} x^T A x &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \\ &= \begin{bmatrix} a_{11}x_1 + a_{21}x_2 & a_{12}x_1 + a_{22}x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \\ &= a_{11}x_1^2 + a_{21}x_1x_2 + a_{12}x_1x_2 + a_{22}x_2^2 \\ &= a_{11}x_1^2 + (a_{21} + a_{12})x_1x_2 + a_{22}x_2^2 \end{aligned}$$

So $x^T A x$ no longer represents a weighted sum of squares.

You can check that the associative law i.e.

$$(x^T A) x = x^T (A x)$$

will apply in both cases (after all, its a law!) as all products are possible.

Exercise 4.5

1.

$$A = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}_{2 \times 3} \quad b = \begin{bmatrix} 9 \\ 6 \\ 0 \end{bmatrix}_{3 \times 1} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1}$$

$$(a) \quad AI = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3} = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} = A$$

$$(b) \quad IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2} \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}$$

$$(c) \quad Ix = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(d) \quad x^T I = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

4. Let's start with a 2×2 diagonal matrix

$$\begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11}^2 & 0 \\ 0 & a_{22}^2 \end{bmatrix}$$

$x = x^2$ for only $x = 0, 1$ so a_{11} and a_{22} can either be 0 or 1. So we can have the following 2×2 idempotent diagonal matrices:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

More generally, for $n \times n$ matrix, there can be 2^n such matrices. This is because there will be n elements, each of which can take two values.

Exercise 4.6

$$2. \quad A = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -8 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}$$

$$(a) \quad A + B = \begin{bmatrix} 3 & -4 \\ -1 & 4 \end{bmatrix}$$

$$A^T + B^T = \begin{bmatrix} 0 & -1 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ -8 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -4 & 4 \end{bmatrix}$$

$$\text{So, } (A + B)^T = A^T + B^T$$

$$(b) \quad AC = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 24 & 4 & 4 \\ 17 & 3 & -6 \end{bmatrix}_{2 \times 3} \quad C^T A^T = \begin{bmatrix} 1 & 6 \\ 0 & 1 \\ 9 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 0 & -1 \\ 4 & 3 \end{bmatrix}_{2 \times 2} =$$

$$\begin{bmatrix} 24 & 17 \\ 4 & 3 \\ 4 & -6 \end{bmatrix}_{3 \times 2} \quad \text{So, } (AC)^T = C^T A^T.$$

Exercise 5.1

3. (a) Yes

(c) Yes

(b) Yes

(d) No; second row = $-2 \times$ first row.

4. Yes, we get the same answer.

Exercise 4.6

6. $A = I - X (X^T X)^{-1} X^T$

(a.) Say the dimension of X is $m \times n$. Then the dimension of $X_{n \times m}^T X_{m \times n}$ is $n \times n$. So the dimension of $(X^T X)^{-1}$ is also $n \times n$. This implies that the dimension of $X_{m \times n} (X^T X)_{n \times n}^{-1} X_{n \times m}^T$ is $m \times m$. Hence, $X^T X$ and A must be square matrices, but X need not be square.

(b.) To prove a matrix is idempotent, we need to show $AA = A$.

$$\begin{aligned}
 AA &= (I - X (X^T X)^{-1} X^T)(I - X (X^T X)^{-1} X^T) \\
 &= I - X (X^T X)^{-1} X^T - X (X^T X)^{-1} X^T + \underbrace{X (X^T X)^{-1} X^T X (X^T X)^{-1} X^T}_I \\
 &= I - X (X^T X)^{-1} X^T - X (X^T X)^{-1} X^T + X (X^T X)^{-1} X^T \\
 &= I - X (X^T X)^{-1} X^T = A
 \end{aligned}$$